DELHI PUBLIC SCHOOL

Bokaro Steel City, Jharkhand

UNIT ASSESSMENT: CBSE GRADE 8 SCIENCE

SUBJECT: CBSE GRADE 8 SCIENCE	**TIME ALLOWED: 2 HOURS**	**MAX MARKS: 30**

Name: ______________________________ Roll No: ______________ Section: ___________

General Instructions:

1. Write your name, roll number, and section clearly in the provided space.
2. All questions are compulsory.
3. Use of calculators is permitted where necessary.
4. Review all answers before final submission.

SECTION A - MULTIPLE CHOICE QUESTIONS

Attempt all questions. Each question carries 1 mark(s).

Q1. Which of the following is the correct unit of electric current? **(1)**

(a) Ampere (A) (b) Volt (V)
(c) Ohm (:'• (d) Watt (W)

Q2. What happens to the resistance of a metallic conductor when its temperature is increased? **(1)**

(a) It increases (b) It decreases
(c) It remains constant (d) It becomes zero

Q3. Three resistors of 3:• each are connected in parallel. What is the equivalent resistance of the circuit? **(1)**

(a) 1:• (b) 3:•
(c) 9:• (d) 0.33:•

Q4. The mathematical expression of Ohm's Law is: **(1)**

(a) V = IR (b) I = VR
(c) R = VI (d) P = VI

SECTION B - SHORT QUESTIONS

Attempt all questions. Each question carries 2 mark(s).

Q1. State Ohm's Law. Mention the conditions under which it holds true. **(2)**

Q2. Define electric potential and potential difference. Write their SI units. **(2)**

Q3. Why are electrical heating elements (like in bread toasters or hair dryers) made of an alloy rather than a pure metal? **(2)**

SECTION C - DIAGRAM/GRAPH-BASED QUESTIONS

Attempt all questions. Each question carries 5 mark(s).

Q1. Draw a circuit diagram representing three resistors in parallel connected to a 12V battery, an ammeter, a voltmeter across one resistor, and a closed key. **(5)**

Q2. Below is a V-I graph plotted for two different metallic wires, A and B. Wire A is steeper than wire B. Which of the wires has higher resistance? Justify your answer. **(5)**

SECTION D - NUMERICAL PROBLEMS

Attempt all questions. Each question carries 5 mark(s).

Q1. An electric bulb is connected to a 220V generator. If the current drawn by the bulb is 0.50A, calculate the power of the bulb and its resistance. **(5)**

Q2. An electric iron consumes energy at a rate of 840W when heating is at the maximum rate. The voltage is 220V. Calculate the current and the resistance. **(5)**